

What wins in professional sports?

The central argument of this project is that conventional wisdom about what makes a basketball team successful is largely wrong. Most fans and even some front offices operate under the assumption that the teams that spend the most money and accumulate the biggest offensive stars win championships. The data suggests otherwise. This project will use NBA team statistics, payroll data, and player efficiency ratings spanning roughly two decades to argue that defensive efficiency and smart roster construction are stronger and more consistent predictors of sustained success than raw spending or scoring. The datasets I plan to use are Basketball Reference team and player statistics (available as clean CSVs going back to 2000), FiveThirtyEight's RAPTOR player ratings and team Elo data (available on GitHub), and NBA payroll data from Spotrac or Basketball Reference.

The core research questions are: How strongly does payroll correlate with wins and playoff success? Is offensive or defensive efficiency a better predictor of championship performance? And which measurable team characteristics most reliably separate playoff teams from non-playoff teams? To answer these, I plan to create a scatter plot of payroll vs. win totals annotated with playoff outcomes, a time-series of offensive vs. defensive efficiency ratings for championship teams over 20 seasons, a correlation heatmap of team statistics against playoff seed, and a bar or lollipop chart ranking teams by defensive efficiency vs. their actual finish to highlight over- and underperformers. Basketball Reference data is extremely clean and well-documented, making it straightforward to work with in Python, and the breadth of available statistics gives me plenty of dimensions to explore across multiple visualization types.

Nutrition Gap

The central argument of this project is that mainstream public health advice around diet — eat more fruits, vegetables, lean proteins, and whole grains — is systemically disconnected from the economic and geographic reality that a large portion of Americans actually live in. Healthy eating is consistently framed as a matter of personal choice, but this project will use county-level food access data and nutritional pricing information to argue that where you live and how much money you make are the dominant constraints on dietary health, not individual decision-making. The datasets I plan to use are the USDA Food Environment Atlas, which provides county-level data on grocery store availability, poverty rates, and diet-related health outcomes like obesity and diabetes prevalence, the USDA FoodData Central database for nutritional profiles and price estimates across food categories, and the Bureau of Labor Statistics Consumer Expenditure Survey for household food spending broken down by income bracket.

The research questions driving the project are: Is there a measurable relationship between county median income and access to fresh produce? Do low-access counties show higher rates of obesity and diabetes? And how does cost-per-nutrient compare across food categories like fresh vegetables, frozen foods, processed snacks, and lean protein? To answer these, I plan to build a choropleth map of food access scores by county colored by income, a scatter plot of food access vs. obesity rate by county, a bar chart comparing price-per-serving across food categories, and a bubble or parallel coordinates chart comparing counties simultaneously across income, food access, health outcomes, and urban/rural classification. The USDA Food Environment Atlas is one of the most comprehensive publicly available datasets on this topic, it is already organized at the county level which makes geographic visualization straightforward,

and it is available as a clean Excel download with clear documentation that should be easy to load and work with in Python.